

FACTSHEET FS11 · Residual momentum

What changed on 2 October 2026. Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS_RERUN_2026-10-02.md.

US, 1999–2012 (pre-registered second period, run 2 October 2026). The same rules on Sharadar's point-in-time S&P 500 with delisted stocks, 1999–2012, a period this factsheet was not built on. L/S net: t -0.12 (2013–2026: +1.22), reversed. Long-only alpha vs the equal-weight universe: +0.1% a year, t +0.05 (2013–2026: +2.30), same sign, weaker. Positive in 2000–2005, negative in 2006–2012 (the 2009 momentum crash). Registered verdicts are unchanged. Pre-registration, method and every result: PREREG_US_1999_2012.md.

Momentum without the market's ups and downs

Henrik Gjerming · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY	ORIGIN	FAMILY
Residual (idiosyncratic) momentum	Blitz, Huij & Martens (2011)	Momentum & trend
UNIVERSES	BACKTEST	BOOKS
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, monthly, net of costs	Long/short and long-only (stocks with the strongest stock-specific trend)
REBALANCE	STATUS	
Monthly	Descriptive factsheet, not a registered trial	

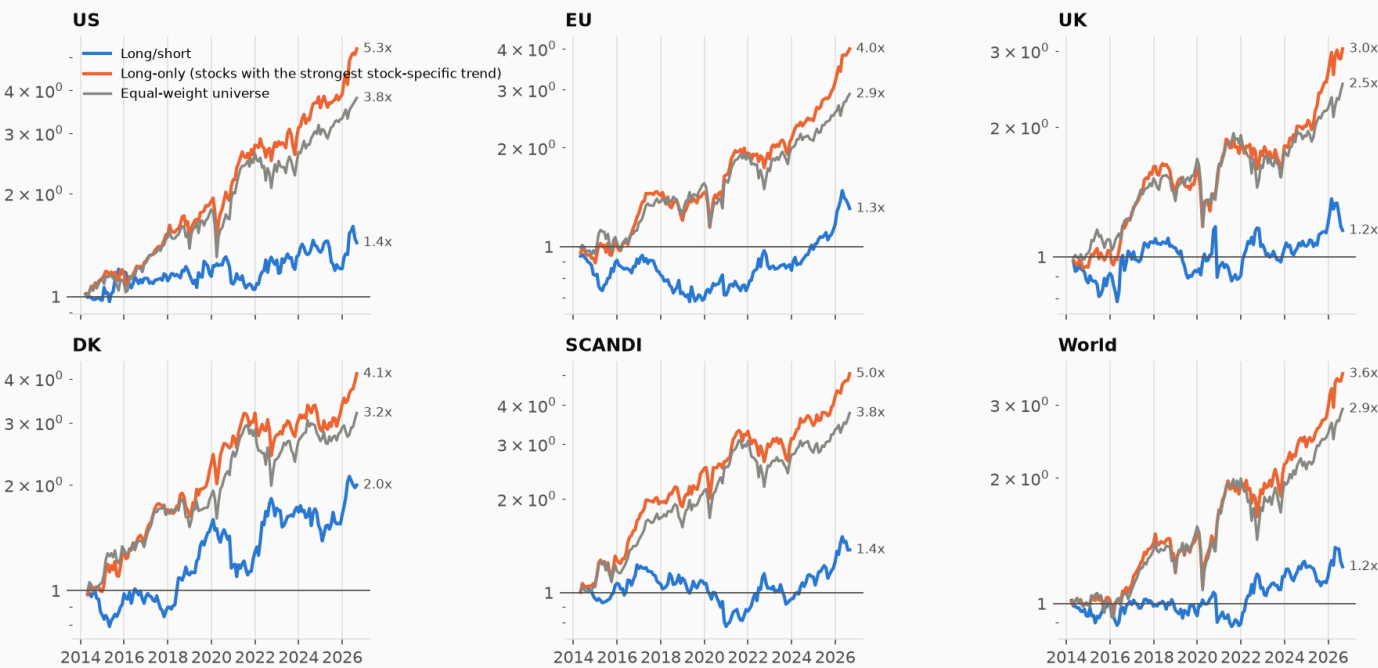
The claim. Ranking stocks on the part of their past return that the market does not explain gives a momentum strategy with about half the volatility and far smaller crashes than plain momentum (Blitz, Huij & Martens 2011, US 1930–2009). **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned +2.2% to +6.9% a year and was positive in all six; no universe passes the 2.87 gate. The long-only book (stocks with the strongest stock-specific trend) had a higher Sharpe than the equal-weight universe in all six, with alphas of +2.4% to +3.9%, none past the gate. **What goes wrong (§10):** the main drags are a market bet (average beta -0.25, worth -2.4% a year in 2013–19), costs of about 2.0% a year. None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Momentum & trend; its returns sit closest to the momentum cluster of the signal library. **360° view (§12):** nearest neighbours Momentum 12-1, Above 52-week low; after its closest neighbours and the market it keeps an alpha of +0.8% to +6.8% (largest |t| 3.5), so it carries some information of its own.

1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
US	499 / 10	+3.8%	1.22	+6.5% (2.2)	-0.22	+14.3%	0.90	-26%	+11.3%	0.76	+14.0%
EU	262 / 10	+2.9%	0.80	+5.7% (1.5)	-0.28	+11.8%	0.86	-22%	+9.0%	0.66	+9.4%
UK	319 / 10	+2.2%	0.63	+4.9% (1.4)	-0.31	+9.3%	0.68	-29%	+7.7%	0.59	+7.6%
DK	20 / 3	+6.9%	1.41	+9.8% (2.2)	-0.28	+12.1%	0.80	-24%	+9.8%	0.69	+7.8%
SCANDI	70 / 5	+3.4%	0.92	+5.8% (1.5)	-0.20	+13.9%	0.95	-21%	+11.3%	0.81	+10.5%
World	1078 / 10	+2.2%	0.83	+4.7% (2.0)	-0.24	+10.7%	0.73	-25%	+9.0%	0.61	+9.4%

Monthly, local currency (World in USD). L/S = equal-weighted stocks with the strongest stock-specific trend minus the weakest, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Residual momentum on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjerner · Project 10 factsheets

2. Strategy description

Ranking stocks on the part of their past return that the market does not explain gives a momentum strategy with about half the volatility and far smaller crashes than plain momentum (Blitz, Huij & Martens 2011, US 1930–2009).

Why it might work. Same under-reaction story as momentum, but removing the market (and factor) component avoids the time-varying beta that makes plain momentum crash (Grundy & Martin 2001; Daniel & Moskowitz 2016).

Rule	Original (Blitz, Huij & Martens (2011))	This factsheet
Signal	Sum of Fama-French 3-factor residuals $t-12..t-2$, over a 36-month window, divided by their volatility	Market-model residuals over 24 months, same scaling
Portfolio	Decile L/S, one month	Extreme quantile L/S, one month
Weighting	Equal weighted	Equal weighted

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP <code>closeadj</code> (total return), 805 tickers incl. delisted	Point-in-time: the S&P 500 members at each month-end (Sharadar add/remove history; corrected 2 Oct 2026)	~503	USD
EU	Project1 <code>yfinance</code> cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** Market-only residuals over 24 months (the paper uses Fama-French 3 factors over 36); stock-level factor loadings outside the US would need data we do not have. The first signals appear in early 2014.

4. Signal creation

$RESMOM_{i,t} = \Sigma \varepsilon_{i,t-11..t-1} / \sigma(\varepsilon)$, from monthly regressions of r_i on the equal-weight universe over the last 24 months. Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).
- **Long/short** = stocks with the strongest stock-specific trend minus the weakest; **long-only** = stocks with the strongest stock-specific trend; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (S&P 500 members, point-in-time), median 499 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+3.8%	+2.9%	13.8%	0.28 [-0.15, 0.66]	0.35	-20%	48%	1.22	-0.22	+6.5% (2.2)	7.6%
Long-only (net)	+14.7%	+14.3%	16.3%	0.90 [0.49, 1.39]	1.46	-26%	64%	4.50	0.92	+3.7% (2.3)	9.3%
Stocks with the strongest stock-specific trend (gross)	+15.8%	+15.5%	16.3%	0.97 [0.55, 1.46]	1.61	-26%	64%	4.85	0.92	+4.8% (3.0)	9.2%
The weakest (gross)	+8.9%	+7.2%	19.5%	0.46 [0.02, 0.98]	0.56	-32%	60%	2.04	1.13	-4.7% (-2.6)	11.9%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
EW universe	+12.0%	+11.3%	15.9%	0.76 [0.32, 1.31]	1.12	-28%	67%	3.64	1.00	–	9.6%
Size-weighted universe	+14.3%	+14.0%	14.5%	0.98 [0.52, 1.55]	1.59	-24%	69%	4.72	0.85	+4.0% (2.5)	8.6%

EU (11 national blue-chip indices, point-in-time), median 262 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.9%	+2.2%	12.6%	0.23 [-0.34, 0.81]	0.25	-28%	55%	0.80	-0.28	+5.7% (1.5)	7.6%
Long-only (net)	+12.2%	+11.8%	14.2%	0.86 [0.36, 1.46]	1.42	-22%	60%	3.22	0.86	+3.9% (1.9)	7.6%
Stocks with the strongest stock- specific trend (gross)	+13.3%	+13.0%	14.2%	0.94 [0.43, 1.54]	1.60	-21%	60%	3.51	0.86	+4.9% (2.4)	7.5%
The weakest (gross)	+7.3%	+5.9%	18.3%	0.40 [-0.05, 0.88]	0.52	-37%	56%	1.62	1.14	-3.7% (-1.7)	10.4%
EW universe	+9.7%	+9.0%	14.6%	0.66 [0.21, 1.19]	0.98	-26%	60%	2.72	1.00	–	8.7%
Size-weighted universe	+10.0%	+9.4%	14.0%	0.72 [0.25, 1.24]	1.09	-25%	58%	2.96	0.91	+1.2% (0.8)	8.1%

UK (FTSE 350, point-in-time), median 319 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.2%	+1.1%	14.6%	0.15 [-0.33, 0.63]	0.11	-25%	52%	0.63	-0.31	+4.9% (1.4)	8.6%
Long-only (net)	+10.1%	+9.3%	14.8%	0.68 [0.18, 1.29]	1.01	-29%	57%	2.65	0.89	+2.5% (1.2)	9.1%
Stocks with the strongest stock- specific trend (gross)	+11.2%	+10.5%	14.8%	0.76 [0.25, 1.37]	1.15	-27%	58%	2.94	0.89	+3.6% (1.7)	9.0%
The weakest (gross)	+6.0%	+4.2%	19.3%	0.31 [-0.14, 0.81]	0.33	-37%	57%	1.28	1.20	-4.2% (-2.2)	12.1%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
EW universe	+8.5%	+7.7%	14.4%	0.59 [0.11, 1.18]	0.80	-30%	58%	2.38	1.00	–	9.2%
Size-weighted universe	+8.2%	+7.6%	13.0%	0.63 [0.12, 1.29]	0.87	-28%	62%	2.43	0.83	+1.2% (0.6)	8.1%

DK (OMXC25, point-in-time), median 20 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+6.9%	+5.7%	16.4%	0.42 [-0.13, 1.02]	0.53	-31%	54%	1.41	-0.28	+9.8% (2.2)	9.7%
Long-only (net)	+12.8%	+12.1%	16.0%	0.80 [0.32, 1.28]	1.25	-24%	58%	3.62	0.89	+3.4% (1.6)	9.1%
Stocks with the strongest stock-specific trend (gross)	+13.4%	+12.8%	16.0%	0.84 [0.36, 1.33]	1.33	-24%	58%	3.81	0.89	+4.0% (1.9)	9.1%
The weakest (gross)	+4.4%	+2.3%	20.4%	0.22 [-0.36, 0.82]	0.16	-56%	58%	0.73	1.16	-7.9% (-2.7)	12.6%
EW universe	+10.6%	+9.8%	15.4%	0.69 [0.18, 1.25]	0.99	-33%	65%	2.70	1.00	–	9.2%
Size-weighted universe	+8.9%	+7.8%	16.6%	0.54 [-0.02, 1.16]	0.69	-40%	59%	2.00	0.93	-0.9% (-0.3)	11.0%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 70 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+3.4%	+2.6%	13.1%	0.26 [-0.30, 0.79]	0.30	-37%	53%	0.92	-0.20	+5.8% (1.5)	7.2%
Long-only (net)	+14.2%	+13.9%	15.0%	0.95 [0.44, 1.53]	1.64	-21%	60%	3.90	0.91	+3.5% (2.0)	7.8%
Stocks with the strongest stock-specific trend (gross)	+15.1%	+15.0%	15.0%	1.01 [0.50, 1.60]	1.79	-20%	60%	4.14	0.91	+4.5% (2.6)	7.7%
The weakest (gross)	+9.2%	+7.9%	17.8%	0.52 [-0.04, 1.11]	0.66	-46%	62%	1.90	1.11	-3.9% (-1.6)	11.5%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
EW universe	+11.8%	+11.3%	14.5%	0.81 [0.28, 1.42]	1.22	-29%	65%	3.16	1.00	–	8.7%
Size-weighted universe	+10.9%	+10.5%	13.9%	0.79 [0.28, 1.35]	1.21	-26%	62%	3.18	0.90	+0.3% (0.2)	8.3%

World (US + UK + EU, point-in-time), median 1078 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.2%	+1.6%	11.1%	0.20 [-0.24, 0.66]	0.22	-17%	51%	0.83	-0.24	+4.7% (2.0)	6.4%
Long-only (net)	+11.4%	+10.7%	15.7%	0.73 [0.27, 1.24]	1.07	-25%	64%	3.23	0.90	+2.4% (1.8)	9.8%
Stocks with the strongest stock-specific trend (gross)	+12.5%	+11.9%	15.7%	0.80 [0.34, 1.32]	1.21	-25%	65%	3.55	0.90	+3.5% (2.6)	9.7%
The weakest (gross)	+7.3%	+5.5%	19.6%	0.37 [-0.09, 0.88]	0.43	-40%	58%	1.53	1.14	-4.1% (-3.1)	11.9%
EW universe	+10.0%	+9.0%	16.3%	0.61 [0.15, 1.16]	0.84	-30%	62%	2.58	1.00	–	10.1%
Size-weighted universe	+10.2%	+9.4%	15.0%	0.68 [0.19, 1.25]	0.98	-28%	61%	2.77	0.90	+1.1% (1.2)	9.2%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2014	-2.5%	-18.3%	-13.0%	-14.8%	-4.5%	-6.8%
2015	+21.1%	+11.3%	+5.2%	+14.9%	+12.1%	+8.1%
2016	-4.1%	-2.2%	+7.8%	-1.9%	+11.5%	-1.3%
2017	-1.8%	-7.9%	+9.1%	-6.7%	-10.7%	-1.0%
2018	+4.1%	-11.4%	-2.6%	+23.5%	+4.4%	-2.1%
2019	-1.0%	-4.6%	-12.1%	+38.5%	-4.2%	-3.0%
2020	-5.0%	+2.7%	+2.4%	-23.1%	-27.0%	-1.4%
2021	-3.5%	+6.3%	-4.2%	+9.0%	+12.0%	-3.0%

Year	US	EU	UK	DK	SCANDI	World
2022	+21.8%	+13.0%	+17.8%	+33.3%	+18.7%	+22.1%
2023	+3.3%	-1.1%	-2.1%	-2.5%	-7.7%	+6.1%
2024	+1.9%	+15.9%	+1.3%	-10.9%	+17.6%	+5.2%
2025	-10.4%	+19.0%	+13.0%	+10.4%	+11.7%	-4.8%
2026	+19.1%	+11.9%	-3.7%	+21.3%	+9.3%	+5.5%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2014	+12.9%	-10.9%	-5.4%	-0.6%	+6.4%	-0.9%
2015	+3.0%	+13.3%	+8.8%	+29.1%	+19.4%	+3.0%
2016	+9.1%	+24.9%	+21.8%	+12.9%	+35.9%	+8.8%
2017	+19.0%	+13.4%	+28.9%	+17.5%	+13.3%	+25.8%
2018	-3.5%	-16.0%	-15.0%	-4.5%	-3.6%	-15.2%
2019	+30.4%	+21.8%	+16.7%	+41.7%	+32.8%	+24.0%
2020	+15.5%	+13.1%	+1.4%	+19.6%	+5.4%	+14.2%
2021	+26.1%	+16.6%	+11.8%	+16.1%	+24.9%	+15.7%
2022	-1.1%	-4.1%	-5.6%	-13.3%	-10.8%	-5.6%
2023	+13.8%	+15.0%	+5.2%	+10.7%	+6.3%	+17.7%
2024	+14.1%	+13.6%	+11.1%	-6.6%	+10.0%	+10.1%
2025	+9.4%	+31.1%	+32.8%	+18.2%	+23.8%	+23.8%
2026	+35.9%	+26.2%	+13.8%	+22.1%	+18.6%	+21.0%

7. Trading record

The rebalance log per universe is in `results/xs_FS11_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	150	50 / 50	46%	46%	48%	-10.2%	2020-11
EU	149	27 / 26	44%	45%	55%	-13.1%	2020-11
UK	149	33 / 32	46%	46%	52%	-23.7%	2020-11
DK	149	8 / 7	28%	28%	54%	-18.9%	2020-11

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
SCANDI	149	15 / 14	38%	37%	53%	-13.7%	2020-11
World	150	109 / 108	46%	46%	51%	-14.2%	2020-11

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: stocks with the strongest stock-specific trend	Short: the weakest
US	HST, PFG, FRT, TGT, IEX, EIX, BIIB, CSX	VST, PLTR, UBER, CSGP, NFLX, META, BSX, TSCO
EU	DANSKE.CO, INVE-B.ST, DHL.DE, MRK.DE, SWED-A.ST, INDU-C.ST, NESTE.HE, LOG.MC	BAR.BR, CTT.LS, AGFB.BR, IVG.MI, RMS.PA, G24.DE, CBK.DE, RHM.DE
UK	KLR.L, EMG.L, MRCH.L, BLND.L, RTW.L, CRDA.L, ROSE.L, LAND.L	RNK.L, ENT.L, RMV.L, STJ.L, HWG.L, AVON.L, FCSS.L, AGT.L
DK	DANSKE.CO, NKT.CO, NOVO-B.CO, CARL-B.CO, JYSK.CO, ORSTED.CO, RBREW.CO, VWS.CO	BAVA.CO, TRYG.CO, MAERSK-A.CO, MAERSK-B.CO, COLO-B.CO, NSIS-B.CO, DSV.CO, AMBU-B.CO
SCANDI	DANSKE.CO, INVE-B.ST, SWED-A.ST, NESTE.HE, INDU-C.ST, ATCO-A.ST, TYRES.HE, JYSK.CO	BAVA.CO, MANTA.HE, KEMIRA.HE, SKF-B.ST, KNEBV.HE, VALMT.HE, TEL2-B.ST, ORNBV.HE

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	+0.7%	0.29	0.54	value -0.29 (-2.3), momentum +0.91 (+8.7), low_risk +0.51 (+3.0), short_term_reversal +0.49 (+2.2)
EU	French Europe 5F + WML	-4.0%	-1.35	0.42	HML +0.40 (+2.4), WML +0.77 (+8.8)
UK	JKP GBR 7 themes	+0.3%	0.11	0.51	momentum +0.95 (+5.4), short_term_reversal +0.41 (+2.3)
DK	JKP DNK 7 themes	+1.0%	0.25	0.34	momentum +0.80 (+5.4), short_term_reversal +0.53 (+3.7)
SCANDI	French Europe 5F + WML	-2.2%	-0.63	0.22	HML +0.33 (+2.2), WML +0.55 (+5.5)
World	JKP World 7 themes	+2.0%	0.63	0.40	value -0.32 (-2.5), momentum +0.65 (+6.8), low_risk +0.30 (+2.3), quality -0.70 (-2.9)

Long-only

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	+3.0%	2.22	0.88	mkt +1.03 (+25.1), momentum +0.39 (+5.8), low_risk +0.32 (+2.6), quality -0.36 (-2.2)
EU	French Europe 5F + WML	+5.2%	2.06	0.62	Mkt-RF +0.71 (+13.1), HML +0.33 (+2.5)
UK	JKP GBR 7 themes	+4.0%	1.94	0.69	mkt +0.55 (+7.8), momentum +0.44 (+5.6), low_risk -0.55 (-2.6), short_term_reversal +0.37 (+2.2)
DK	JKP DNK 7 themes	+5.6%	2.22	0.59	mkt +0.67 (+9.1), size -0.29 (-2.0), short_term_reversal +0.28 (+2.4)
SCANDI	French Europe 5F + WML	+7.6%	2.14	0.49	Mkt-RF +0.67 (+8.5)
World	JKP World 7 themes	+5.7%	2.61	0.80	mkt +0.71 (+13.5), momentum +0.22 (+2.5)

9. Statistical verdict

- **Gate:** $|t| > 2.87$ (2 books × 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in all six; passes in 0 of 6.
- **L/S CAPM alpha:** +4.7% to +9.8%, $|t|$ past the gate in 0 of 6.
- **Long-only:** Sharpe above the equal-weight universe in all six; alpha +2.4% to +3.9%, passing in 0 of 6.
- **Probabilistic Sharpe** of the L/S, $P(\text{true Sharpe} > 0)$: US 0.84, EU 0.79, UK 0.70, DK 0.93, SCANDI 0.82, World 0.76.
- **Publication decay:** gross L/S -2.6% to +10.5% a year in 2013–19 against +7.5% to +13.3% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FSO3–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FSO3–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	+6.1%	-0.41	-4.4%	+10.5%	-2.2%	+2.0%	+4.1%	91%	2015-04 -6%, 2016-02 -6%, 2019-11 -5%
EU	-2.6%	-0.34	-2.9%	+0.2%	-2.1%	-0.1%	-2.6%	89%	2018-11 -9%, 2015-02 -8%, 2014-04 -7%

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
UK	+2.5%	-0.33	-3.2%	+5.7%	-2.2%	+0.3%	+2.1%	94%	2016-08 -8%, 2015-06 -6%, 2019-04 -5%
DK	+10.5%	-0.06	-0.7%	+11.2%	-1.4%	+4.6%	+5.9%	59%	2014-11 -8%, 2016-04 -7%, 2016-10 -5%
SCANDI	+4.2%	-0.05	-0.8%	+4.9%	-1.8%	+3.8%	+0.3%	73%	2018-11 -6%, 2018-07 -6%, 2016-04 -5%
World	+2.4%	-0.33	-2.6%	+5.0%	-2.3%	+0.4%	+2.0%	94%	2016-02 -6%, 2019-04 -5%, 2015-02 -4%

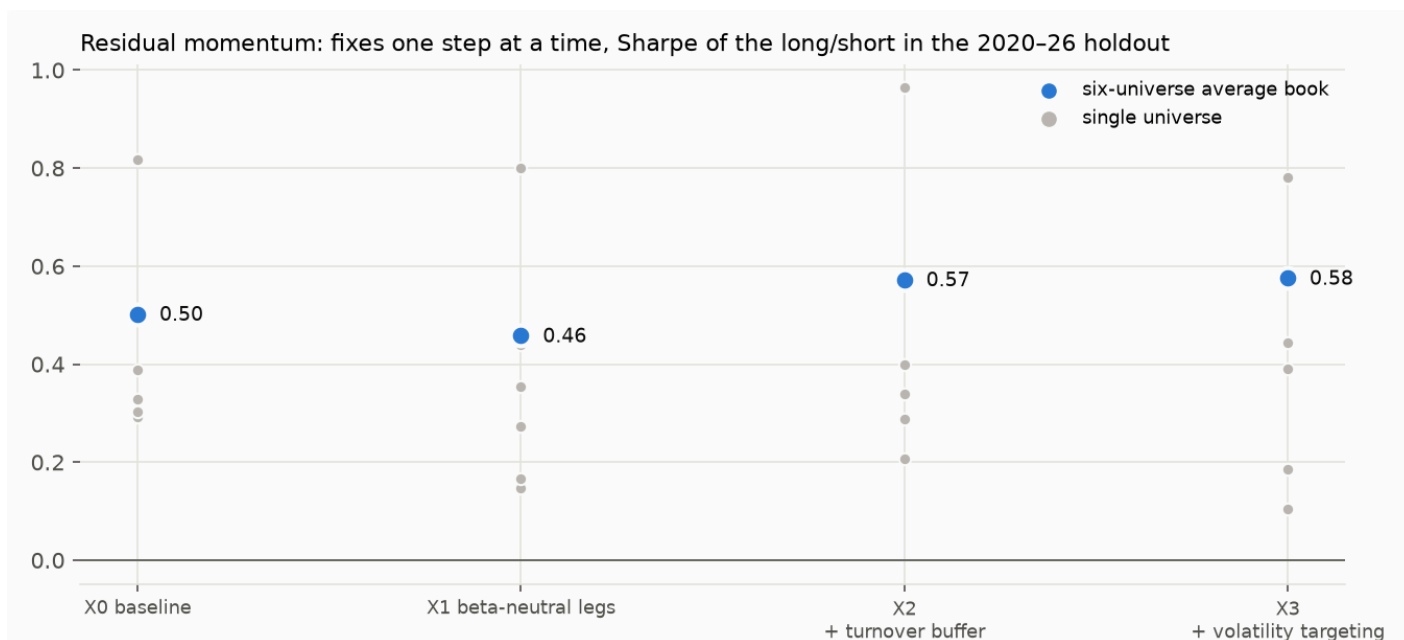
On average across the six universes the gross spread was +3.8% a year, of which the market exposure (beta -0.25) contributed -2.4%; the beta-adjusted spread (CAPM alpha) was +6.3%. Costs took 2.0% a year at 83% monthly turnover across both legs. The long leg beat the universe by +1.8% and the short leg lagged it by +2.0% a year, so most of the spread comes from the short side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +6.8%, +4.4%, +4.2% a year.

10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	0.08	–	0.50	–	–	+5.8%	-16%	+8.9% (2.4)	2.0%
X1 beta-neutral legs	0.12	+0.09 (p 0.202)	0.46	-0.04 (p 0.618)	2 of 6	+4.5%	-18%	+6.2% (1.6)	2.1%
X2 + turnover buffer	0.14	+0.04 (p 0.387)	0.57	+0.11 (p 0.063)	6 of 6	+4.9%	-13%	+6.4% (2.0)	1.1%
X3 + volatility targeting	0.07	-0.10 (p 0.593)	0.58	+0.00 (p 0.495)	1 of 6	+5.7%	-17%	+7.1% (2.0)	1.0%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



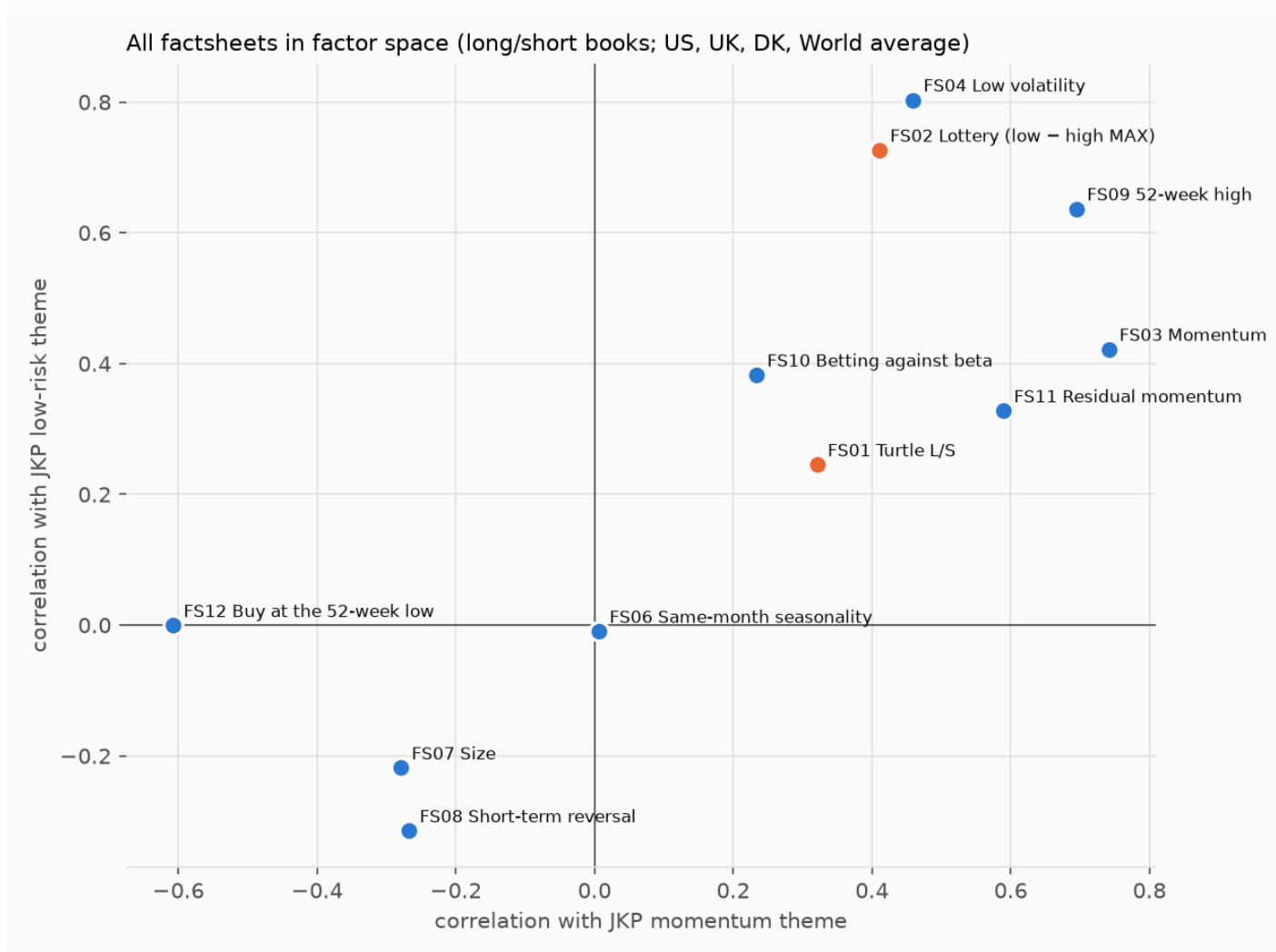
- **X1 beta-neutral legs:** +0.09 design, -0.04 holdout; helps in one window and hurts in the other: not reliable.
- **X2 + turnover buffer:** +0.04 design, +0.11 holdout; helps in both windows but does not pass the gate.
- **X3 + volatility targeting:** -0.10 design, +0.00 holdout; helps in one window and hurts in the other: not reliable.

Where it ends: holdout Sharpe 0.50 → 0.58, return +5.8% → +5.7% a year (six-universe book).

11. Classification: where does it fit?

Dimension	Residual momentum
Series family	Momentum & trend
Academic style	Residual (idiosyncratic) momentum
Signal	Price only (daily total returns)
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 83% of the two legs replaced monthly
Market exposure	L/S beta -0.31 to -0.20
Payoff shape	Treynor–Mazuy γ -1.66 to -0.25 (t -3.4 to -0.2); worst 10% of market months +0.7% to +3.5% a month, best 10% -2.6% to -0.9%
Nearest factor theme	momentum (4 of 6 universes), WML (2 of 6 universes)
Nearest library signals (returns)	Momentum 12-1 (6 universes), Near 52-week high (6 universes), Above 52-week low (5 universes)
Economic rationale	Same under-reaction story as momentum, but removing the market (and factor) component avoids the time-varying beta that makes plain momentum crash (Grundy & Martin 2001; Daniel & Moskowitz 2016).

Dimension	Residual momentum
Publication	Grundy & Martin 2001; Gutierrez & Pirinsky 2007; Blitz, Huij & Martens 2011
Where it fits	In the momentum cluster of the library; with some information beyond its neighbours.

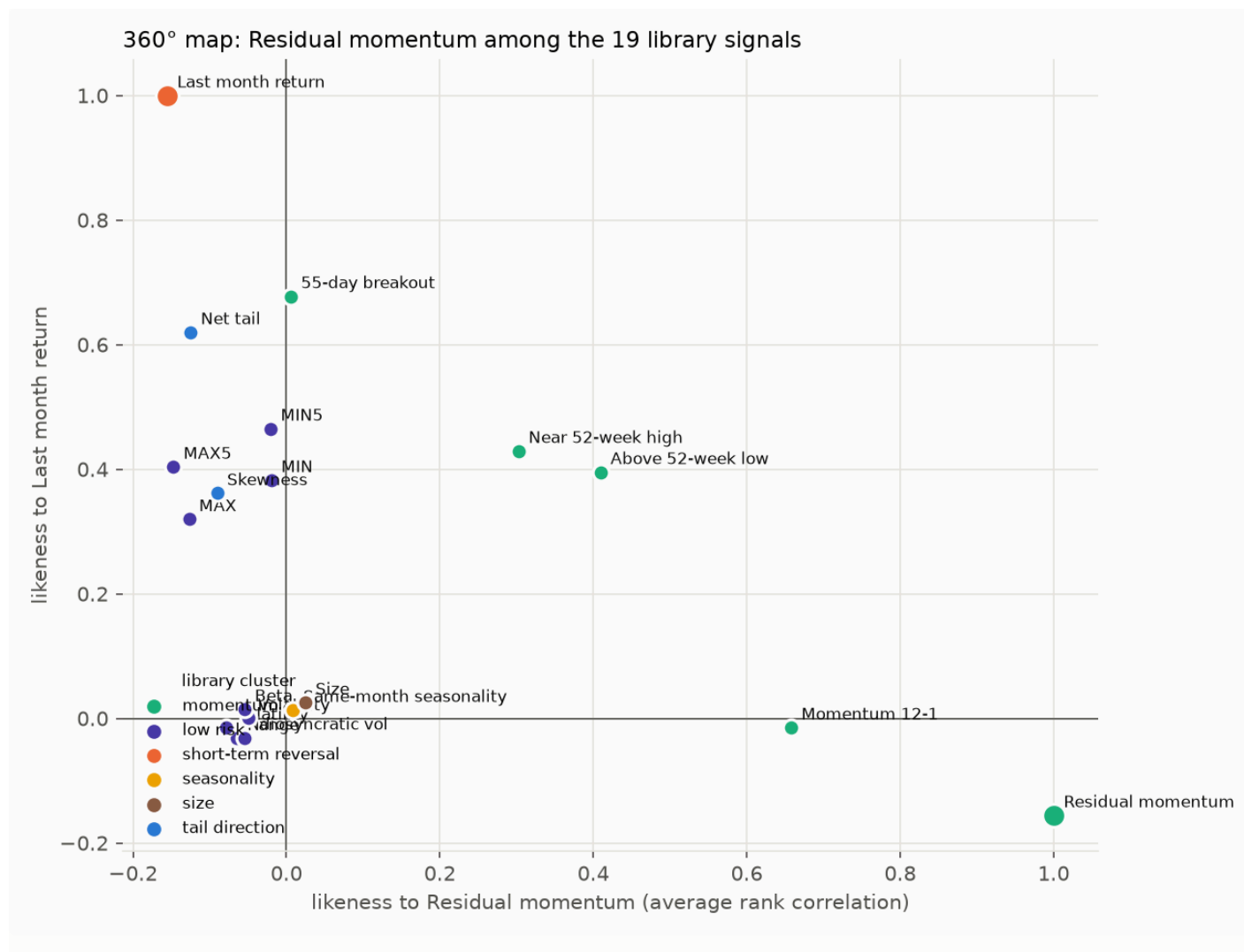


12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Last month return**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha $t \geq +2 / \leq -2$
Residual momentum (this strategy)	momentum	+1.00	+1.00	+6.4%	+8.0% to +12.2%	6 / 0
Momentum 12-1	momentum	+0.66	+0.64	+7.2%	+4.3% to +19.4%	5 / 0
Near 52-week high	momentum	+0.30	+0.46	+3.7%	+9.3% to +22.3%	6 / 0
Above 52-week low	momentum	+0.41	+0.44	+7.5%	+5.2% to +13.6%	4 / 0

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha $t \geq +2 / \leq -2$
MAX5	low risk	-0.15	-0.31	+0.6%	-11.3% to -3.5%	0 / 2
Volatility	low risk	-0.08	-0.31	+1.1%	-13.7% to -7.9%	0 / 5
MAX (best day)	low risk	-0.13	-0.29	+0.1%	-13.3% to -2.0%	0 / 2
Range (MAX–MIN)	low risk	-0.06	-0.29	-0.7%	-15.4% to -4.8%	0 / 4
Beta	low risk	-0.05	-0.28	+4.4%	-12.6% to -4.2%	0 / 4
Volatility (252 days)	low risk	-0.05	-0.26	+2.4%	-12.9% to -4.8%	0 / 3
55-day breakout	momentum	+0.01	+0.24	+1.4%	+4.1% to +13.0%	4 / 0
MIN5	low risk	-0.02	+0.24	-0.9%	+2.3% to +13.7%	4 / 0
Idiosyncratic vol	low risk	-0.05	-0.23	-3.6%	-15.5% to -4.8%	0 / 4
MIN (worst day)	low risk	-0.02	+0.23	-0.7%	+3.3% to +11.1%	4 / 0
Skewness	tail direction	-0.09	-0.13	-0.4%	-6.7% to +3.9%	0 / 2
Size	size	+0.02	+0.11	+0.1%	-3.8% to +9.9%	1 / 0
Net tail (MAX+MIN)	tail direction	-0.12	-0.07	-0.9%	-5.2% to +5.2%	0 / 0
Same-month seasonality	seasonality	+0.01	-0.03	-1.9%	-9.4% to +6.6%	0 / 1
Last month return (mirror)	short-term reversal	-0.16	+0.01	+1.8%	-3.2% to +11.0%	2 / 0



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

Universe	Alpha after neighbours / yr	t	R ²	Neighbours with t ≥ 2
US	+6.8%	3.54	0.51	Momentum 12-1 +0.39 (+3.3), Near 52-week high +0.31 (+2.2), Size -0.30 (-2.8), 55-day breakout -0.36 (-3.6)
EU	+0.9%	0.29	0.42	Momentum 12-1 +0.42 (+4.4)
UK	+1.8%	0.59	0.51	Momentum 12-1 +0.39 (+3.9), Above 52-week low +0.22 (+2.4)
DK	+6.0%	1.65	0.41	Momentum 12-1 +0.62 (+8.2), MAX5 -0.18 (-2.2)
SCANDI	+4.5%	1.50	0.41	Momentum 12-1 +0.51 (+5.6), Above 52-week low +0.33 (+3.4), Near 52-week high -0.34 (-2.9)
World	+0.8%	0.37	0.61	Momentum 12-1 +0.38 (+4.0), Above 52-week low +0.31 (+3.6), 55-day breakout -0.39 (-4.0), Volatility -0.24 (-4.1)

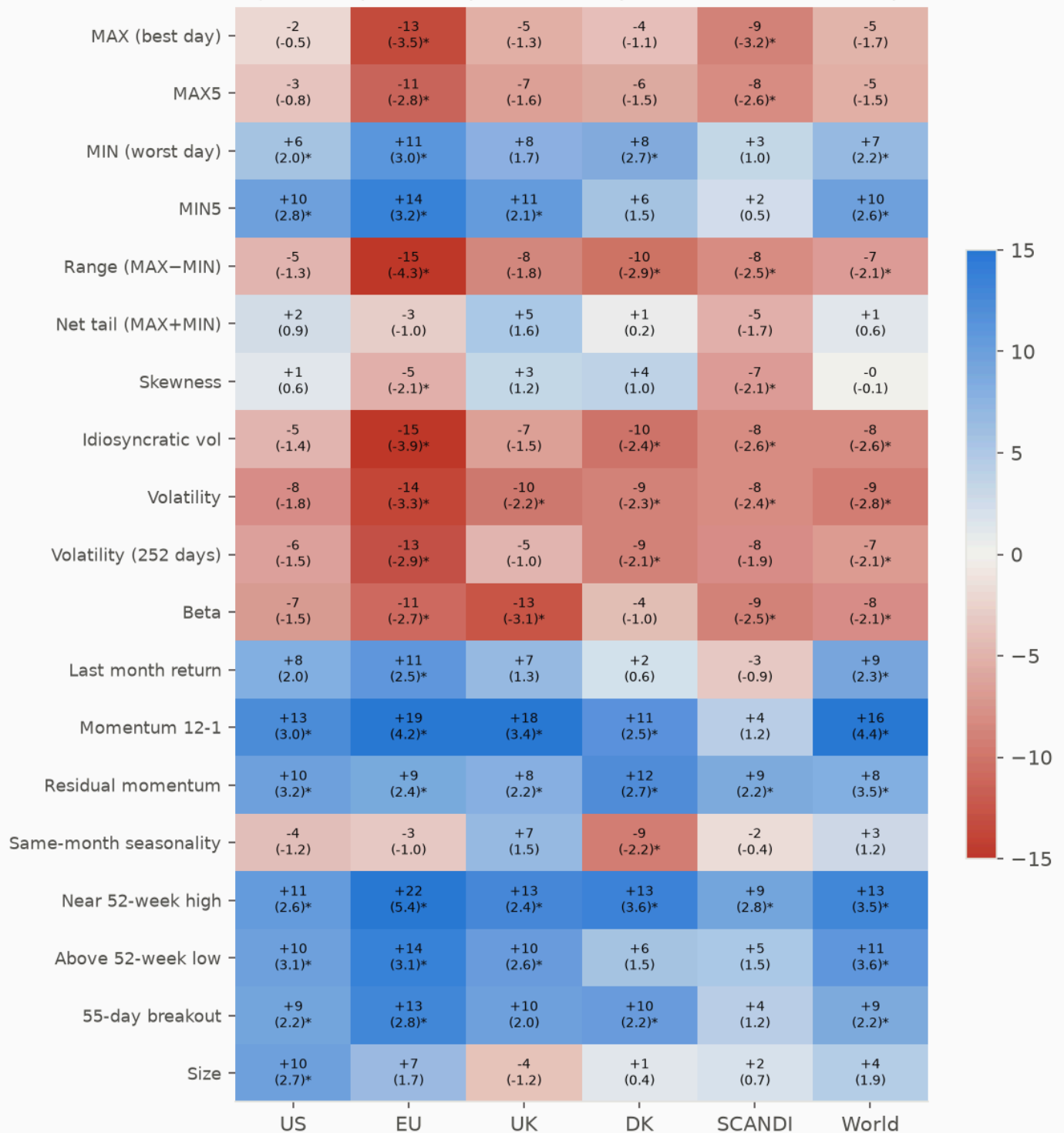
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Residual momentum -	7.8%	7.9%	9.0%
middle -	11.8%	11.0%	9.5%
high Residual momentum -	12.2%	12.4%	12.3%
	low Last month return	middle	high Last month return

Holding Last month return fixed (down a column), moving from low to high Residual momentum changes the return by +4.4, +4.5 and +3.3 points a year; holding Residual momentum fixed (along a row), moving from low to high Last month return changes it by +1.2, -2.3 and +0.0. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -1% vs +19% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +6.4%, +6.8%, +4.4%, +4.2% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.

3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.
4. **Data gap.** Market-only residuals over 24 months (the paper uses Fama-French 3 factors over 36); stock-level factor loadings outside the US would need data we do not have. The first signals appear in early 2014.
5. **Not a registered trial.** Descriptive; the fix ladder is counted inside FSO3–FS12 (25 trials).

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15. Reproduce

code/data.py → signals.py, signals2.py, library.py → xs.py + analyse_xs.py FS11 → build_xs.py FS11.

Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.